Kshitij Kanade

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SUMMARY

Electrical & Computer Engineering graduate student at UC San Diego with a diverse technical background spanning industry, research, personal projects, and technical leadership across hardware and software domains. Experience ranges from embedded and real-time systems, cybersecurity, robotics, and computational engineering to machine learning, parallel computing, and system automation, complemented by project management and cross-functional leadership. Strong foundation in C/C++, Python, Verilog/VHDL, and system-level problem solving, with a track record of learning across disciplines and translating ideas into working systems. Seeking opportunities to build on this broad foundation while developing deep specialization in a high-impact technical domain.

EDUCATION

University of California, San Diego — La Jolla, California, US
Master of Science, Electrical and Computer Engineering, **Track: Computer Engineering** — June 2028
Relevant Coursework: VLSI Digital System Algorithms and Architectures, Principles of Computer Architecture, Optimization and Acceleration of Deep Learning on Various Hardware Platforms, Low Power VLSI Implementation for Machine Learning.

National Institute of Technology, Tiruchirappalli — Trichy, India
Bachelor of Technology, Electrical and Electronics Engineering, **GPA: 9.13/10** — July 2025
Relevant Coursework: Data Structures and Algorithms, Machine Learning and Deep Learning, Big Data Analytics, Web Technology, Ethical Hacking, Networks and Linear Systems, Power System Analysis, Control Systems, Digital Signal Processing, Power Electronics, Linear Integrated Circuits, Digital System Design and HDLS, VLSI Design, Linear Algebra and Calculus, Complex Analysis and Differential Equations, Fourier Transforms and Numerical Techniques

PERSONAL PROJECTS

Parallel Computation of AXPY | **[GitHub Repo]** — Nagpur/Aug 2026 – Ongoing

- Engineered a C++ parallel-computing workflow spanning serial CPU execution, OpenMP multithreading, SIMD/vectorization analysis, and CUDA GPU execution using the AXPY kernel (`y = αx + y`).
- Benchmarked CPU and parallel implementations through repeated timing runs, statistical analysis, outlier detection, and memory-bandwidth calculations to systematically evaluate execution performance.
- Built and validated a Windows acceleration toolchain using MSVC x64, OpenMP, CUDA 13.3/nvcc, and VS Code; successfully compiled and executed CUDA kernels on an NVIDIA RTX 3050 Laptop GPU.
- Developed reusable benchmarking infrastructure to separate performance measurement and statistical analysis from computational kernels, by maintaining the project in a version-controlled GitHub repository for reproducibility and future cross-platform extension.

Automated Maintenance Script — Nagpur/June 2026 – Aug 2026

- Engineered a privileged, logon-triggered Windows maintenance automation system using PowerShell, Task Scheduler, and AutoHotkey, orchestrating system updates, application updates, security scans, drive optimization, and temporary-file cleanup through a configurable interactive workflow.
- Designed a state-driven maintenance pipeline with persistent execution logs and 24-hour scheduling logic, dynamically determining maintenance status and presenting user-controlled task selection, execution, restart/shutdown, or deferral workflows.
- Built resilient automation around unpredictable and GUI-dependent operations, implementing polling, configurable timeouts, retries, cancellation handling, graceful/forced termination, and failure-state tracking across individual maintenance modules.
- Developed a reusable interactive orchestration layer supporting dynamic task selection, Select All, countdown controls, keyboard interruption, elevated execution, visible/hidden process modes, and multiple post-execution system states.
- Performed 1,000+ iterative test/debug cycles over ~4 weeks, systematically exercising edge cases, interruption paths, timing variations, and failure scenarios to stabilize the automation and preserve execution state across subsequent logon sessions.

WORK EXPERIENCE

Modelling and Control of Velocity Skin Effect and Plasmas Observed in EM Railguns, IIT Bombay | **[Cert]** — Mumbai, India
Project Technical Assistant under Dr. Himanshu J. Bahirat, Electrical Dept., IIT Bombay — Feb 2026 – May 2026

- Built 3+ modular FEM solvers in Python (Poisson, 1D transmission line, sparse-matrix FEM), enabling rapid experimentation with EM diffusion and current distribution.
- Evaluated 15+ COMSOL solver configurations (segregated, fully coupled, direct, iterative) and 10+ meshing strategies, improving stability in EC–MF coupled simulations by ~40% through refined continuity and identity-pair handling.
- Diagnosed 20+ convergence failures (mesh-induced instability, tolerance mismatch, ill-conditioning, NaN/Inf states) and implemented corrective workflows that reduced solver divergence across complex rail geometries.
- Conducted 30+ electromagnetic field analyses (current density, magnetic flux, skin-effect penetration), generating validated insights for next-generation localized railgun FEM solvers.
- Benchmarked 6 sparse solvers (LU, GMRES, CG, BiCGStab, PARDISO variants), reducing memory overflow cases by ~25% through optimized matrix assembly and mesh partitioning.

- Improved mesh quality near curved conductor regions by ~35%, eliminating inverted elements and enhancing field accuracy in high-gradient zones.

Vajra Cybersecurity Lab, A91 Eco Hub, TCAAI, IIT Bombay | [Cert] Mumbai, India
Senior Research Intern under Dr. Manjesh Kumar Hanawal, IEOR Dept., IIT Bombay Dec 2024 – June 2025

- Contributed to a cross-platform cybersecurity platform (VAJRA EDR) for detecting malicious files and processes and executing response actions.
- Converted 1000+ C++ rules to optimized SQL queries, boosting detection efficiency by 9.5% and cutting backend alert runtime from 30 min to <7 s.
- Automated MITRE Caldera workflows with Python and Selenium, reducing validation time 10 min to 1 <3 min.
- Resolved 20+ critical frontend/backend issues and mentored 6 interns on the MITRE ATT&CK framework and system automation.

Centre for Excellence in Robotics, IIT Kharagpur | [Cert] Kharagpur, India
Summer Research Intern under Dr. D. K. Pratihar, Mech. Dept., IIT Kharagpur May 2024 – July 2024

- Upgraded an Educational Doll with touch classification ("Good", "Bad", "May Not Like It") to a wireless system (98% accuracy at 5 cm range).
- Enhanced an Agricultural Robot (Patent #550470) with YOLOv8 and Mask R-CNN, attaining 95% navigation accuracy.
- Integrated Intel RealSense D455 and force sensors, reducing inference latency from 500 ms to 80 ms and enabling a 1.5 N auto-stop grip.
- Improved fruit detection and picking precision by 8% through optimized vision and control algorithms for a 6-DOF robotic arm.

Leadership Experience

PSI Racing Club – ***DAQ & E-Powertrain Head / DAQ Engineer*** Trichy/July 2023 - July 2025

- Led DAQ and e-powertrain development for SAEINDIA mBAJA/eBAJA vehicles, progressing from DAQ Engineer to DAQ & E-Powertrain Head and owning sensor integration, telemetry, and vehicle-level electrical systems.
- Architected the embedded sensing and telemetry pipeline, integrating sensors with Arduino/MicroPython controllers, processing noisy real-world signals, and transmitting live vehicle data to the pit crew over Zigbee for real-time monitoring and performance analysis.
- Designed and integrated HV/GLV systems for the eBAJA platform, contributing to selection and integration of the motor, controller, accumulator, and BMS, while developing instrumentation for battery and electrical-system monitoring.
- Developed ML-based battery state-of-charge, thermal, and performance prediction models and integrated insights into the telemetry workflow; contributed to AIR 2 nationally in mBAJA Phase 1, AIR 5 in Phase 2, AIR 19 in Phase 3, and AIR 2 among eBAJA New Teams.

Electrical and Electronics Engineer's Association – ***Vice Chairperson / Treasurer / Manager*** Trichy/Sept 2023 - July 2025

- Progressed through OC, Events (Manager) → Treasurer → Vice Chairperson, coordinating cross-functional teams and overseeing technical, cultural, and departmental initiatives.
- Managed a ₹2.78L organizational budget, handling allocations, procurement, sponsorships, and expenditure tracking while reducing budget variance by 7.3%.
- Led new-event development and execution across planning, logistics, volunteers, and stakeholder coordination, contributing to a 15% increase in participation.

Maximus—The Math Society of NITT—***Vice President & Overall Coordinator/ Manager (Events)*** Trichy/Sept 2023 - July 2025

- Progressed from Events Manager to Vice President & Overall Coordinator, overseeing society operations, flagship events, and strategic initiatives.
- Directed and mentored 20+ coordinators across planning, logistics, marketing, partnerships, and execution for 6+ flagship events.
- Introduced interactive and interdisciplinary event formats while managing external collaborations and event operations, contributing to 2.5% higher participation and 5% improved event satisfaction.

Omega Science & Spirituality Club – ***Treasurer / Events Manager*** Trichy/Sept 2023 - July 2025

- Progressed from Organizing Committee/Events to Treasurer, coordinating seminars, workshops, and meditation initiatives serving 1,800+ participants.
- Managed a ₹27K budget across event planning, procurement, and sponsorship coordination; outreach initiatives contributed to 40% higher participation and a reported 5× increase in engagement.

Skills and Interests

Languages: C/C++, Python, Verilog/VHDL, SQL, MATLAB, Java, MicroPython, TypeScript
Hardware & Systems: COMSOL Multiphysics, ANSYS, CREO, LTSpice, OpenMP, Git, Linux/Windows Toolchains
Numerical & ML Frameworks: TensorFlow, Keras, OpenCV, PySparse, PARDISO, GMRES/CG/BiCGStab Solvers, NumPy,SciPy
Data & Web: Hadoop, Streamlit, Tableau, MongoDB, Postman, HTML/CSS
Soft Skills: Project Management, Technical Documentation, Research Communication, Problem-Solving, Teamwork, Leadership

Certifications: Amazon ML Summer School 2024, Machine Learning Specialization (Stanford Online, Coursera 2023), Python for Everybody (University of Michigan, Coursera 2023), Data Analytics with Python (NPTEL), Adhyayan's ACA Summer School 2023 (IIT Kanpur - Deep Learning for Computer Vision, Natural Language Processing), Introduction to Embedded Machine Learning (Edge Impulse, Coursera 2026)